CDAC ADDMISSION PROCESS

CREATING TABLES

```
KD3_Rahil_96207>CREATE TABLE ccat_result(
    ->     formno INT PRIMARY KEY,
    ->     name VARCHAR(50) NOT NULL,
    ->     ab_rank INT UNIQUE
    -> );
Query OK, 0 rows affected (0.977 sec)

KD3_Rahil_96207>CREATE TABLE centers(
    ->     centerid INT PRIMARY KEY,
    ->     cname VARCHAR(50),
    ->     caddress VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.575 sec)

KD3_Rahil_96207>CREATE TABLE courses(
    ->     courseid INT PRIMARY KEY,
    ->     courname VARCHAR(50),
    ->     cfees INT CHECK(cfees > 0)
    -> );
Query OK, 0 rows affected (0.311 sec)

KD3_Rahil_96207>CREATE TABLE center_courses(
    ->     centerid INT,
    ->     courseid INT,
    ->     capacity INT CHECK(capacity > 0),
    ->
    ->     PRIMARY KEY(centerid, courseid),
    ->
    ->     FOREIGN KEY(centerid) REFERENCES centers(centerid),
    ->     FOREIGN KEY(courseid) REFERENCES courses(courseid)
    -> );
Query OK, 0 rows affected (0.493 sec)
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```
KD3_Rahil_96207>CREATE TABLE preferences(
    ->     formno INT,
    ->     pref_no INT,
    ->     centerid INT,
    ->     courseid INT,
    ->     freeze CHAR(1) DEFAULT 'N',
    ->
    ->     PRIMARY KEY(formno, pref_no),
    ->
    ->     FOREIGN KEY(formno) REFERENCES ccat_result(formno),
    ->     FOREIGN KEY(centerid) REFERENCES centers(centerid),
    ->     FOREIGN KEY(courseid) REFERENCES courses(courseid)
    -> );
Query OK, 0 rows affected (0.545 sec)

KD3_Rahil_96207>CREATE TABLE allocations(
    ->     formno INT,
    ->     centerid INT,
    ->     courseid INT,
    ->     round_no INT,
    ->
    ->     PRIMARY KEY(formno, round_no),
    ->
    ->     FOREIGN KEY(formno) REFERENCES ccat_result(formno)
    -> );
Query OK, 0 rows affected (0.532 sec)

KD3_Rahil_96207>INSERT INTO ccat_result VALUES
    -> (101,'Aman',3),
    -> (102,'Riya',1),
    -> (103,'Karan',5),
    -> (104,'Neha',2),
    -> (105,'Arjun',4);
Query OK, 5 rows affected (0.227 sec)
Records: 5  Duplicates: 0  Warnings: 0

KD3_Rahil_96207>INSERT INTO centers VALUES
    -> (1,'Pune','Pune'),
    -> (2,'Mumbai','Mumbai');
Query OK, 2 rows affected (0.174 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
KD3_Rahil_96207>INSERT INTO courses VALUES
    -> (1,'DAC',90000),
    -> (2,'DBDA',85000);
Query OK, 2 rows affected (0.176 sec)
Records: 2  Duplicates: 0  Warnings: 0

KD3_Rahil_96207>INSERT INTO center_courses VALUES
    -> (1,1,2),
    -> (1,2,1),
    -> (2,1,1),
    -> (2,2,1);
Query OK, 4 rows affected (0.052 sec)
Records: 4  Duplicates: 0  Warnings: 0

KD3_Rahil_96207>INSERT INTO preferences VALUES
    -> (101,1,1,1,'N'),
    -> (101,2,1,2,'N'),
    -> (101,3,2,1,'N'),
    -> (101,4,2,2,'N'),
    -> (101,5,1,1,'N');
Query OK, 5 rows affected (0.051 sec)
Records: 5  Duplicates: 0  Warnings: 0

KD3_Rahil_96207>INSERT INTO preferences VALUES
    -> (102,1,1,1,'N'),
    -> (102,2,2,1,'N'),
    -> (102,3,1,2,'N'),
    -> (102,4,2,2,'N'),
    -> (102,5,1,1,'N');
Query OK, 5 rows affected (0.047 sec)
Records: 5  Duplicates: 0  Warnings: 0

KD3_Rahil_96207>INSERT INTO preferences VALUES
    -> (103,1,2,1,'N'),
    -> (103,2,1,1,'N'),
    -> (103,3,2,2,'N'),
    -> (103,4,1,2,'N'),
    -> (103,5,1,1,'N');
Query OK, 5 rows affected (0.165 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
KD3_Rahil_96207>INSERT INTO preferences VALUES
    -> (104,1,1,1,'N'),
    -> (104,2,2,1,'N'),
    -> (104,3,1,2,'N'),
    -> (104,4,2,2,'N'),
    -> (104,5,1,1,'N');
Query OK, 5 rows affected (0.191 sec)
Records: 5  Duplicates: 0  Warnings: 0

KD3_Rahil_96207>INSERT INTO preferences VALUES
    -> (105,1,1,2,'N'),
    -> (105,2,2,1,'N'),
    -> (105,3,1,1,'N'),
    -> (105,4,2,2,'N'),
    -> (105,5,1,1,'N');
Query OK, 5 rows affected (0.164 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

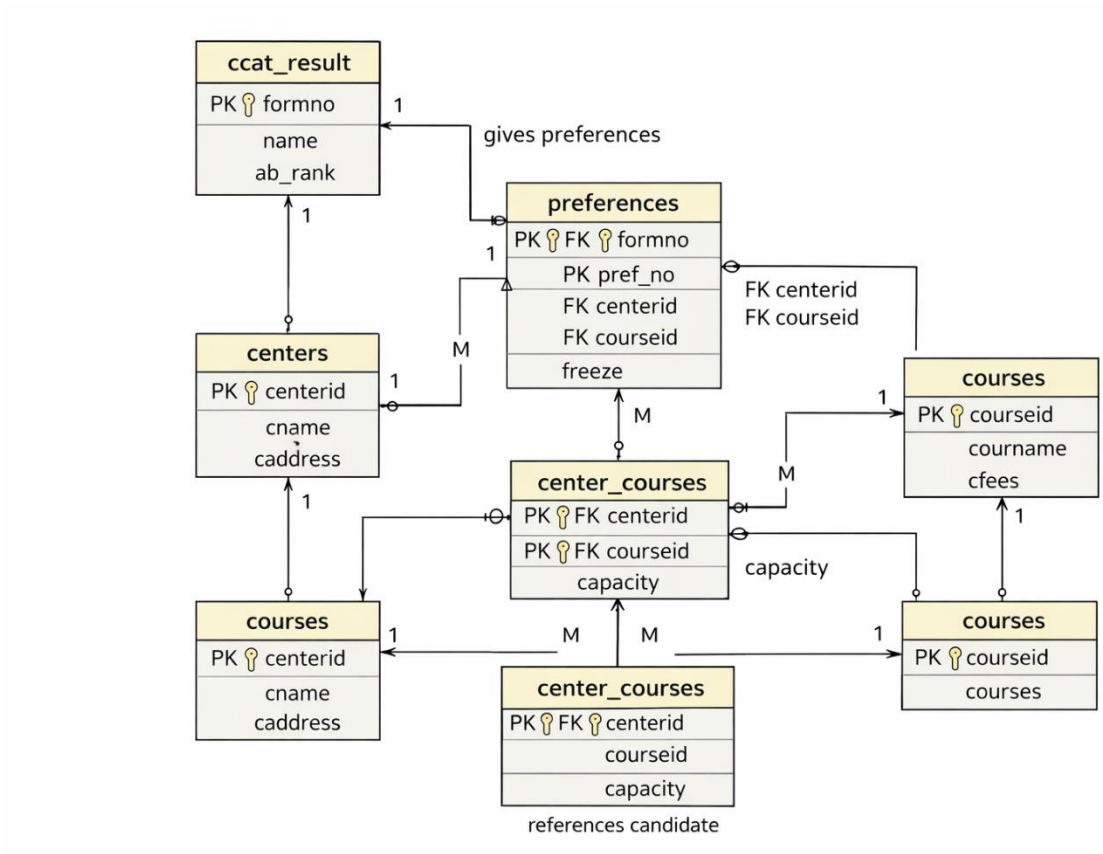
----------STORED PROCEDURE USING CURSOR ----------

```
KD3_Rahil_96207>CREATE PROCEDURE round1_allocation()
    -> BEGIN
    ->
    -> DECLARE done INT DEFAULT 0;
    -> DECLARE v_formno INT;
    ->
    -> DECLARE cur_candidates CURSOR FOR
    ->     SELECT formno FROM ccat_result ORDER BY ab_rank;
    ->
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    -> OPEN cur_candidates;
    ->
    -> candidate_loop: LOOP
    ->
    -> FETCH cur_candidates INTO v_formno;
    ->
    -> IF done = 1 THEN
    ->     LEAVE candidate_loop;
    -> END IF;
    ->
    -> INSERT INTO allocations(formno, centerid, courseid, round_no)
    ->
    -> SELECT p.formno, p.centerid, p.courseid, 1
    -> FROM preferences p
    -> JOIN center_courses cc
    -> ON p.centerid = cc.centerid AND p.courseid = cc.courseid
    ->
    -> WHERE p.formno = v_formno
    ->
    -> AND (
    ->     SELECT COUNT(*)
    ->     FROM allocations a
    ->     WHERE a.centerid = p.centerid
    ->     AND a.courseid = p.courseid
    -> ) < cc.capacity
    ->
    -> ORDER BY p.pref_no
    -> LIMIT 1;
    ->
    -> END LOOP;
    ->
    -> CLOSE cur_candidates;
    ->
```

RESULT

```
KD3_Rahil_96207>CALL round1_allocation();
Query OK, 0 rows affected (0.322 sec)

KD3_Rahil_96207>SELECT * FROM allocations;
+--------+----------+----------+----------+
| formno | centerid | courseid | round_no |
+--------+----------+----------+----------+
|    101 |        1 |        2 |        1 |
|    102 |        1 |        1 |        1 |
|    103 |        2 |        2 |        1 |
|    104 |        1 |        1 |        1 |
|    105 |        2 |        1 |        1 |
+--------+----------+----------+----------+
5 rows in set (0.005 sec)

KD3_Rahil_96207>
```

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